

Fourier analysis – 3

The Fourier transform on the line

Homework

13 сентября 2026 года

Theoretical minimum. The transform exchanges differentiation with multiplication by the variable, convolution with pointwise product, and translation with modulation; hence smoothness on one side is decay on the other. On the Schwartz class it is a bijection with the explicit inverse, and Plancherel's identity makes it unitary on L^2 . The Gaussian is a fixed point, and Gaussian kernels form an approximate identity, which is how one proves inversion in the first place. Poisson summation converts a sum over the integers into a sum over the dual lattice and produces the functional equation of the theta function. Two rigidity facts are worth remembering: a function and its transform cannot both be concentrated (the uncertainty principle, with equality only for Gaussians), and they cannot both have compact support.

1. (S.T. Yau CSMC 2018, team) Let $\widehat{f}(\xi) = \int e^{-ix\xi} f(x) dx$ for f in the Schwartz class $\mathcal{S}(\mathbb{R})$. Suppose that $f(2\pi n) = 0$ and $\widehat{f}(n) = 0$ for all integers n . Prove that $f = 0$.
2. (S.T. Yau CSMC 2016) Let f be integrable on $\mathbb{R}^n$ and put $K_\delta(x) = \delta^{-n/2} e^{-\pi|x|^2/\delta}$. Prove that $f * K_\delta$ is infinitely differentiable and that $f * K_\delta \rightarrow f$ in L^1 as $\delta \rightarrow 0^+$.
3. (S.T. Yau CSMC) A function $\varphi: \mathbb{R}^n \rightarrow \mathbb{C}$ is positive definite if $\sum_{i,j=1}^k c_i \overline{c_j} \varphi(y_i - y_j) \geq 0$ for all k , all $y_j \in \mathbb{R}^n$ and all $c_j \in \mathbb{C}$. For a measurable positive definite φ prove that
 - a) $\varphi(-y) = \overline{\varphi(y)}$ and $|\varphi(y)| \leq \varphi(0)$;
 - b) $\int \int \varphi(x - y) f(x) f(y) dx dy \geq 0$ for every non-negative integrable f ;

- c) $\int \varphi(x)e^{-\alpha|x|^2} dx \geq 0$ for every $\alpha > 0$.
4. Prove that the Fourier transform of an integrable function is uniformly continuous and tends to zero at infinity, and construct a continuous function vanishing at infinity which is not the transform of any integrable function.
 5. Prove that if f is integrable and $\hat{f} \geq 0$ with f continuous at 0, then $\hat{f}$ is integrable and the inversion formula holds pointwise.
 6. Prove the Heisenberg inequality in the form $\|f\|_2^2 \leq 2\|xf\|_2\|f'\|_2$ by integrating $\frac{d}{dx}(x|f|^2)$ by parts, and explain how it is related to problem 4 of the class sheet.
 7. Compute the Fourier transforms of $\frac{\sin ax}{x}$, of $e^{-a|x|}$ and of $\frac{1}{a^2+x^2}$, and use Plancherel's identity to evaluate $\int_{-\infty}^{\infty} \frac{dx}{(a^2+x^2)^2}$.
 8. Prove that the Fourier transform maps the Schwartz class into itself bijectively, and that it does not map L^1 into L^1 .
 9. Using Poisson summation, prove that $\sum_{n \in \mathbb{Z}} \frac{1}{n^2+a^2} = \frac{\pi}{a} \coth(\pi a)$ and compare with the residue proof of Complex analysis – 4.
 10. Prove the Paley–Wiener statement in one direction: if f is supported in $[-A, A]$ and integrable, then $\hat{f}$ extends to an entire function of exponential type A ; and deduce once more that f and $\hat{f}$ cannot both have compact support.